Aravind Reddy Talla

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Research Interests

- *Theoretical Machine Learning:* Beyond-worst-case analysis, clustering, probabilistic inference, streaming and dynamic algorithms, and diverse subset selection (DPPs).
- *Responsible AI:* Multilingual and multicultural AI safety evaluation; fair clustering and machine unlearning.
- *Computer Vision:* Data-efficient and multimodal learning.

Education

Northwestern University — Evanston, Illinois, USA
Ph.D. in Computer Science; Advisors: Konstantin Makarychev and Aravindan Vijayaraghavan — Jun 2023
M.S. in Computer Science; GPA: 4.0/4.0 — Mar 2021

Indian Institute of Technology (IIT) Kanpur — May 2018
Bachelor of Technology (B.Tech.) in CSE; minor in Physics; Distinction

Academic and Research Appointments

Principal Project Scientist, Centre for Responsible AI (CeRAI), IIT Madras — 3 Nov 2025–present

Visiting Assistant Professor, Department of AI, IIT Hyderabad — 22 Jul 2024–21 Jul 2025

Postdoctoral Associate, Broad Institute of MIT and Harvard — 5 Jul 2023–10 Jan 2024
Deep learning for genomic structural variant discovery. Mentor: Victoria Popic — Cambridge, MA, USA

Research Intern, Adobe Research — Summers 2022 and 2021
2022 mentor: Ritwik Sinha; 2021 mentors: Ryan Rossi and Zhao Song. — San Jose, CA, USA (remote)

Research Intern, Université Paris Diderot — Summer 2017
Paris Centre for Quantum Technologies; mentors: Eleni Diamanti and Iordanis Kerenidis. — Paris, France

Research Intern, National University of Singapore — Summer 2016
Centre for Quantum Technologies; mentor: Hartmut Klauck. — Singapore

Research Grants

Google Research Gift (US$100,000) — 1 Oct 2026–31 Mar 2028
Role: Co-authored the proposal with Dr. Lora Aroyo (Google DeepMind); project lead at CeRAI.
Project title: *Scaling Pluralis to Enable Multilingual AI Safety Evaluation for Agents Across the Globe.*
PI: Prof. Balaraman Ravindran, IIT Madras; **sanctioned cost:** INR 95,90,910.

Adobe Research Gift (US$20,000) — 2022
Role: Co-authored the proposal with Dr. Zhao Song and Dr. Ritwik Sinha (Adobe Research).
Project title: *Dimension Reduction to Enable Learning on the Edge.*
PI: Prof. Aravindan Vijayaraghavan, Northwestern University.

Awards & Honours

Silver Reviewer Award, ICML 2026 Position Paper Track — 2026

Peter and Adrienne Barris Outstanding TA Award, Northwestern University; press release — 2022

CS Supplemental Award ($3,000), Northwestern; top CS Ph.D. applicants — 2018

Charpak Scholarship for Research Internship, Government of France — 2017

Academic Excellence Award, IIT Kanpur; top 10% by first-year CPI — 2015

National Initiative on Undergraduate Science Scholarship, HBCSE (TIFR) — 2015

All India Rank 189, JEE Advanced — 2014

Gold Medallist, Indian National Physics Olympiad; national top 35 — 2014

Publications

Eight peer-reviewed conference papers (four CORE A^*, two CORE A; seven published, one accepted at NeurIPS 2026); one granted US patent.

[FA] First author [CF] Co-first author [AO] Alphabetical author order
Conference ranks use the ICORE2026 (formerly CORE) Conference Ranking.

1. X-AVDD: Cross-Attentive Audio-Visual Dataset Distillation
Saumyaranjan Mohanty, **Aravind Reddy**, Konda Reddy Mopuri — *NeurIPS 2026 (CORE A*); arXiv*

2. DiRe: Diversity-promoting Regularization for Dataset Condensation
Saumyaranjan Mohanty, **Aravind Reddy**, Konda Reddy Mopuri — *WACV 2026 (CORE A)*

3. Online Adaptive Mahalanobis Distance Estimation[AO]
Lianke Qin, **Aravind Reddy**, Zhao Song — *IEEE BigData 2023 (CORE B)*

4. Adaptive and Dynamic Multi-Resolution Hashing for Pairwise Summations[AO]
Lianke Qin, **Aravind Reddy**, Zhao Song, Zhaozhuo Xu, Danyang Zhuo — *IEEE BigData 2022 (CORE B)*

5. Dynamic Tensor Product Regression[AO]
Aravind Reddy, Zhao Song, Lichen Zhang — *NeurIPS 2022 (CORE A*)*

6. One-Pass Algorithms for MAP Inference of Nonsymmetric Determinantal Point Processes[FA]
Aravind Reddy, Ryan A. Rossi, Zhao Song, and others — *ICML 2022 (CORE A*)*

7. Beyond Perturbation Stability: LP Recovery Guarantees for MAP Inference on Noisy Stable Instances[CF]
Hunter Lang, **Aravind Reddy**, David Sontag, Aravindan Vijayaraghavan — *AISTATS 2021 (CORE A)*

8. Improved Guarantees for k-means++ and k-means++ Parallel[AO]
Konstantin Makarychev, **Aravind Reddy**, Liren Shan — *NeurIPS 2020 (CORE A*)*

Granted US Patent

Online Inference and Learning for Nonsymmetric Determinantal Point Processes — Apr. 29, 2025
Ryan A. Rossi, **Aravind Reddy Talla**, Zhao Song, and others — *US12288237B2*

Preprints / Papers Under Review

[*] Equal contribution.

1. Machine Unlearning for Federated Socially Fair k-Means Clustering[CF]
Aravind Reddy[*], Deeraj S K[*], Mohd Wasif Raza[*], Saurav Prakash — *arXiv*

2. Pluralis v0.1: Towards a Multicultural, Multimodal, Multilingual Benchmark for AI Risk and Reliability
with S. Sitaram, B. Ravindran, L. Aroyo, and others; **India lead** — *arXiv*

Teaching Experience

Instructor, IIT Hyderabad

Convex Optimization / Optimization-I — AI 2101 / AI 1233
Jan–May 2025 · Full semester · 158 enrolled; 124 respondents
Instructor rating: **4.02/5** · Course rating: **4.03/5**

Programming for AI — AI 1013
Jan–Apr 2025 · 10 weeks · 36 enrolled; 34 respondents
Co-taught with Dr. P. N. Karthik
Individual instructor rating: **4.58/5** · Course rating: **4.71/5**

Introduction to Modern Artificial Intelligence — AI 1001
Sep–Oct 2024 · 5 weeks · 103 enrolled; 90 respondents
Instructor rating: **4.02/5** · Course rating: **4.07/5**

Introduction to Submodular Functions — EE 5328
Jul–Sep 2024 · 5 weeks · 5 enrolled; 4 respondents
Instructor rating: **4.70/5** · Course rating: **4.35/5**

Teaching Assistant

Northwestern University

Mathematical Foundations of Computer Science (CS 212): Spring 2022, Fall 2020, Fall 2019.

Received the Peter and Adrienne Barris Outstanding Teaching Assistant Award for Spring 2022.

Design & Analysis of Algorithms (CS 336): Winter 2019.

IIT Kanpur Winter 2018

Fundamentals of Computing (ESC 101).

Guest Lectures

Probability and Probabilistic Modelling; Linear Algebra Sep–Oct 2026

Mathematical Foundations for Machine Learning, AI Training Programme for the Indian Navy (two lectures).

Maximum Likelihood and Probabilistic Linear Regression Aug 2026

Machine Learning I, IIT Madras (one week, three lectures).

Submodular Function Maximization Apr 2022

Algorithms for Big Data, Northwestern University.

Introduction to Markov Chains May 2019

Math Toolkit for TCS, Northwestern University.

Volunteer Math Tutor

Tutoring Chicago, 2021–2022: Tutored a Chicago-area middle-school student in middle- and high-school mathematics.

Research Mentoring

Formal: project mentor, WSAI Summer Internship Program, IIT Madras (faculty guide: Prof. B. Ravindran)

Madhangi Karimanal, B.E. (CSE), SSN College of Engineering, Chennai May–Jul 2026

Project: *AILuminate: Multicultural, Multimodal and Multilingual Benchmark for AI Risk and Reliability.* Co-author of Pluralis v0.1.

Paruchuri Sriram, M.Tech. (CSE), IIIT Hyderabad May–Jul 2026

Project: *Saakshya: A Multi-Agent, RAG-Enhanced Framework for Deepfake Detection and Governance.*

Informal research mentoring

Saumyaranjan Mohanty, Ph.D. student, IIT Hyderabad

Advisor: Dr. Konda Reddy Mopuri. Joint papers: DiRe (WACV 2026) and X-AVDD (NeurIPS 2026).

Mohd Wasif Raza, M.S. student (EE), IIT Madras

Co-first author, *Machine Unlearning for Federated Socially Fair k-Means Clustering.*

Deeraj S K, B.Tech. student, SVNIT Surat

WSAI summer intern (faculty guide: Dr. Saurav Prakash). Co-first author, *Machine Unlearning for Federated Socially Fair k-Means Clustering.*

Akanksha Singh, Post-baccalaureate Fellow, WSAI, IIT Madras

Faculty guide: Prof. B. Ravindran.

Vraj Thakkar, Post-baccalaureate Fellow, WSAI, IIT Madras Ongoing

Faculty guide: Dr. Krishna Pillutla.

Service and Outreach

Conference Reviewer: AAAI (2026); AISTATS (2023–2024); ECCV (2026); FSTTCS (2024); ICLR (2022–2024); ICML (2022–2026); NeurIPS (2021–2025).

Invited Speaker: "How to Teach AI When Labels Are Expensive", for school students at the Research Science Initiative Chennai (RSI-C) (May 2026); for 300 high-school students at Oak Valley International School, Hyderabad (Jun 2025).

Workshop Leader and Mentor: NeurIPS 2022 High-School Outreach Program.

Board Member: CS Ph.D. Advisory Council, Northwestern University (2020–2022).

Webmaster: Northwestern CS Theory group (2018–2023).

Student Volunteer: STOC (2020, online); FSTTCS (2017, IIT Kanpur).

Coordinator, IIT Kanpur student groups: Association for Computing Activities (ACA), Card & Board Games Club (CBG), and Science Coffeehouse (SCH) (2016–2017). Matched freshmen with senior student mentors for semester projects through ACA. Helped organize Microsoft code.fun.do at IIT Kanpur (2017), contributing to scheduling and local logistics.

Invited Research Talks

On MAP Inference of Ferromagnetic Potts Models and Nonsymmetric DPPs
IIT Madras, Department of Data Science & AI — Sep 2024
IIT Kanpur, Department of Computer Science & Engineering — Jun 2024
IISc Bangalore, Bangalore Theory Seminar; recording — May 2024
IIT Hyderabad, Department of CSE — May 2024

On Clustering, MAP Inference, and Causal Inference
Broad Institute of MIT and Harvard, Popic Lab — Nov 2022
Stanford University, Aghaeepour Lab — Oct 2022

Beyond Perturbation Stability: LP Recovery Guarantees for MAP Inference on Noisy Stable Instances University of Chicago Theory Lunch — Nov 2022

Conference Presentations and Posters

Dynamic Tensor Product Regression NeurIPS, New Orleans — Dec 2022

One-Pass Algorithms for MAP Inference of Nonsymmetric Determinantal Point Processes
NSF TRIPODS meeting (poster), virtual — Sep 2022
ICML, Baltimore — Jul 2022

Beyond Perturbation Stability: LP Recovery Guarantees for MAP Inference on Noisy Stable Instances AISTATS, virtual — Apr 2021

Improved Guarantees for k-means++ and k-means++ Parallel NeurIPS, virtual — Dec 2020

Conference Travel Grants

NeurIPS Scholar Award ($2,000) 2022

FOCS Travel Award ($425) 2022

ICML Participation Grant ($400) 2022

STOC Student Travel Grant ($600) 2019

Languages

English, Telugu, Hindi (native/bilingual proficiency); Korean (limited working proficiency).

Personal Interests

Running: Half-marathon finisher, NMDC Hyderabad Marathon, 2026 (2:34:02).